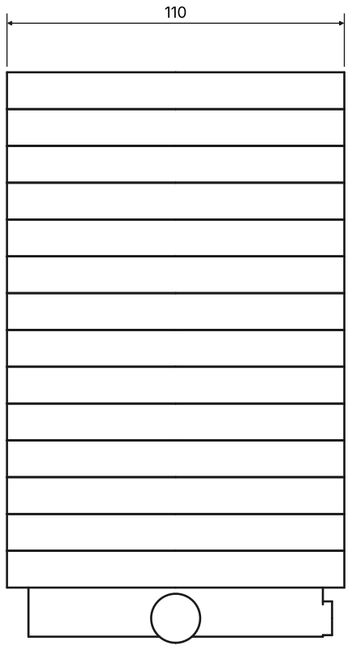
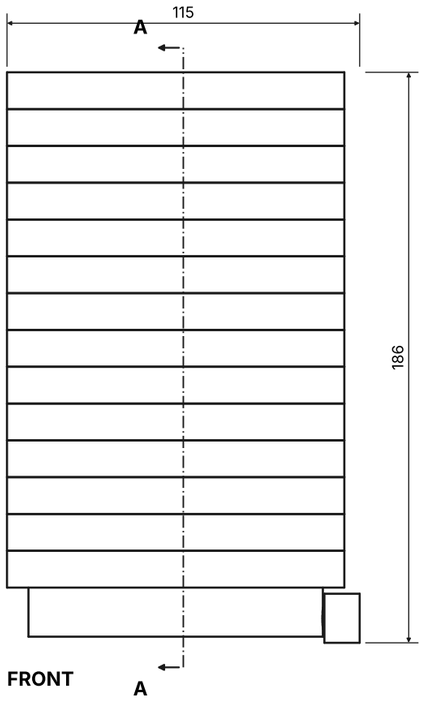
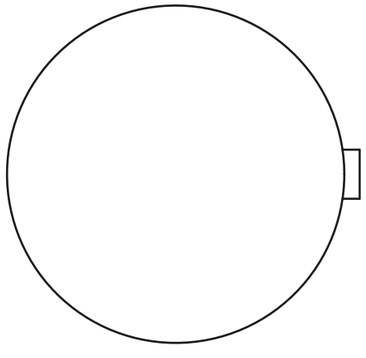
Scale 1:2. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

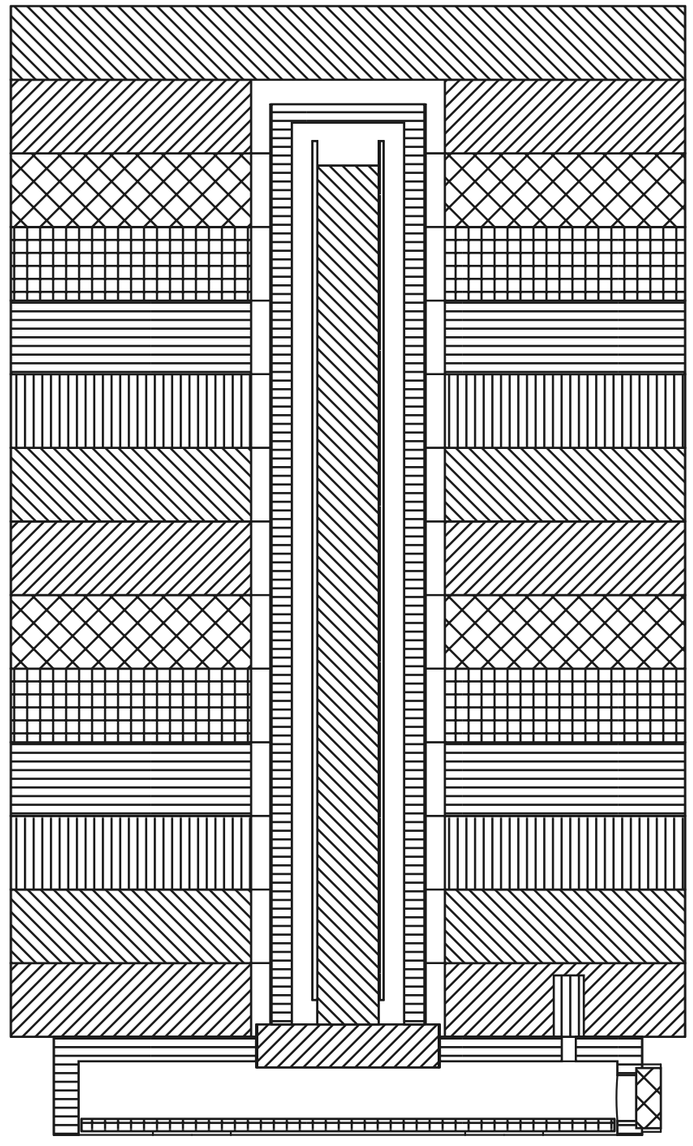
General arrangement

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. Section A-A (sheet 1): the product cut at x 2.5 and seen from the right; cut faces hatched, a different angle for each part.



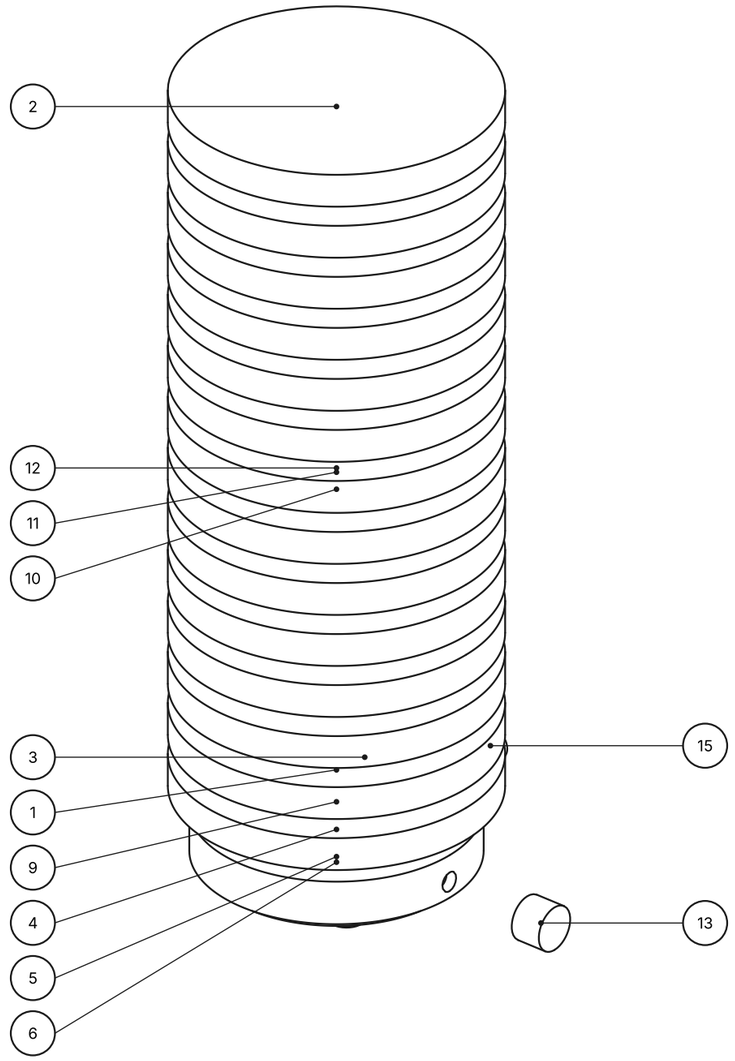
Section A-A

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2, parts drawn apart. Balloons are the parts list's numbers (last sheet).



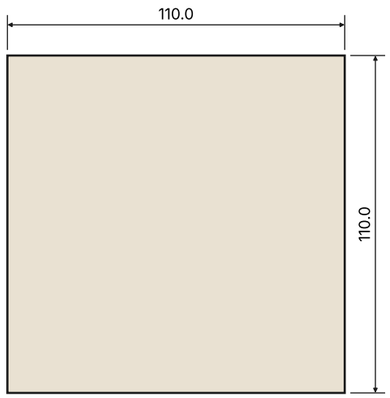
Exploded view

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

mm | issue 2026-10-31 | first angle

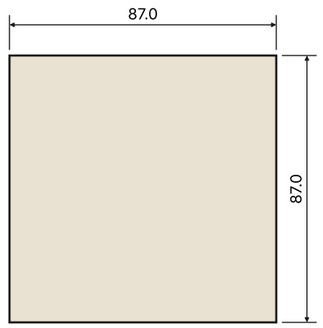
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



disc-14

12 mm cast-acrylic, 1 for the set



cover

2 mm aluminium, 1 for the set

Sheet parts: plain panels

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

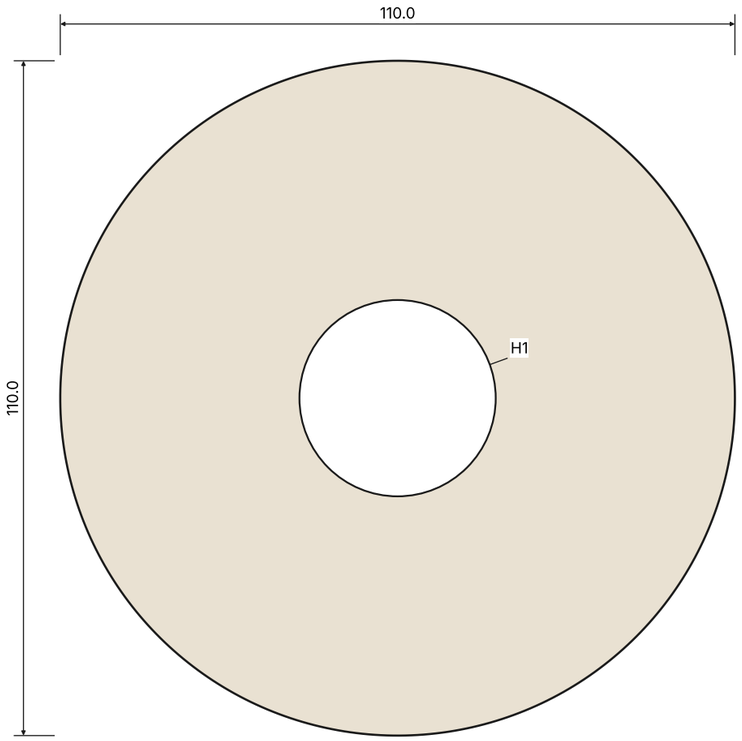
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-01.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-01: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-01

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

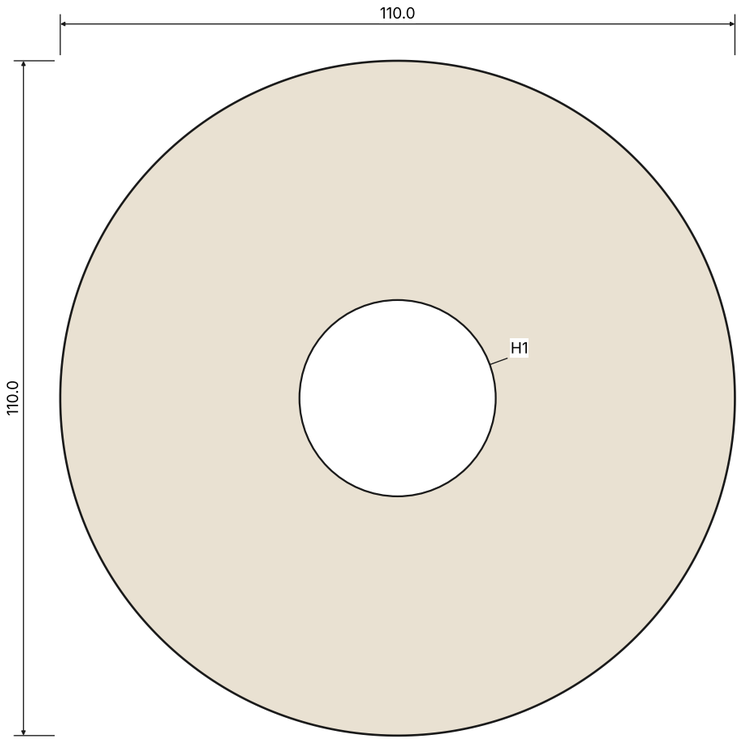
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-02.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-02: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-02

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

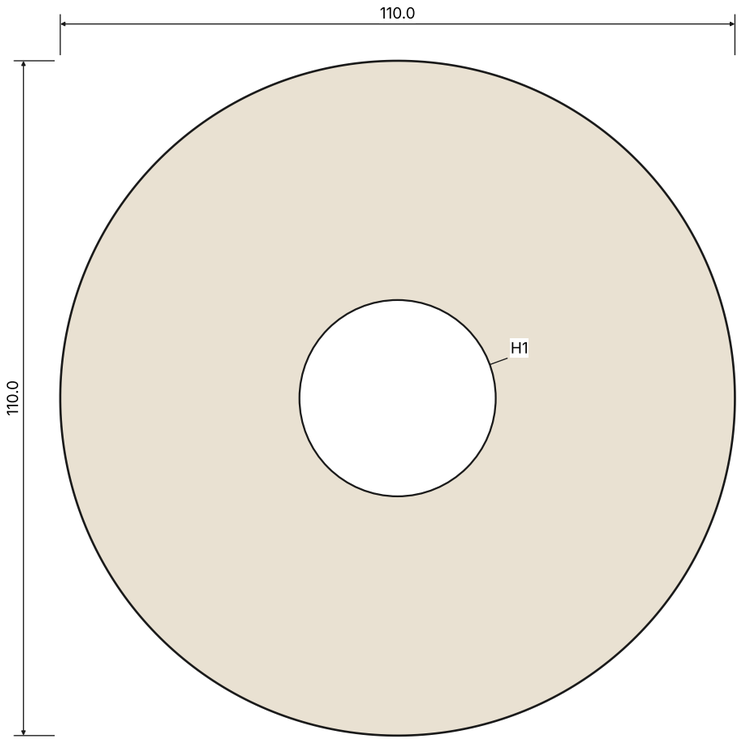
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-03.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-03: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-03

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

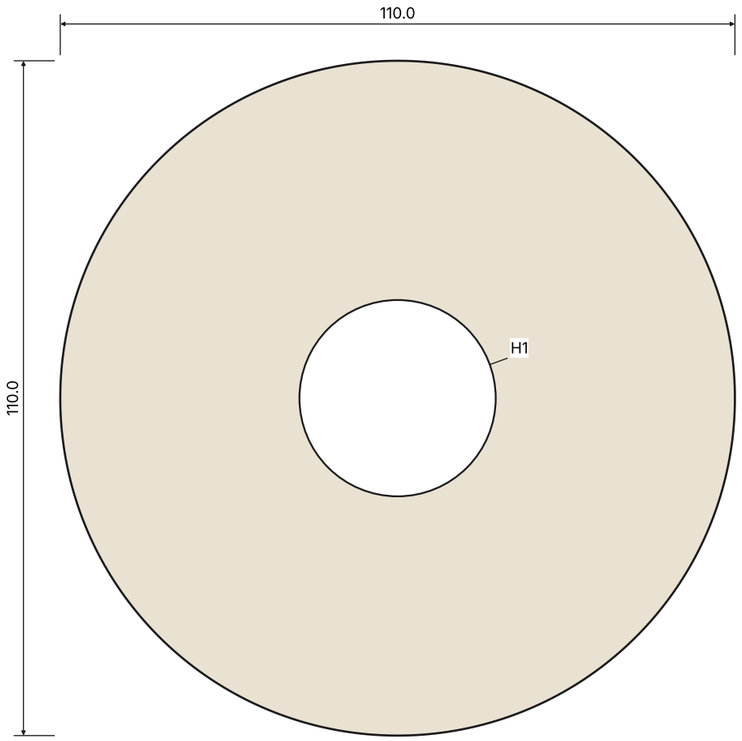
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-04.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-04: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-04

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

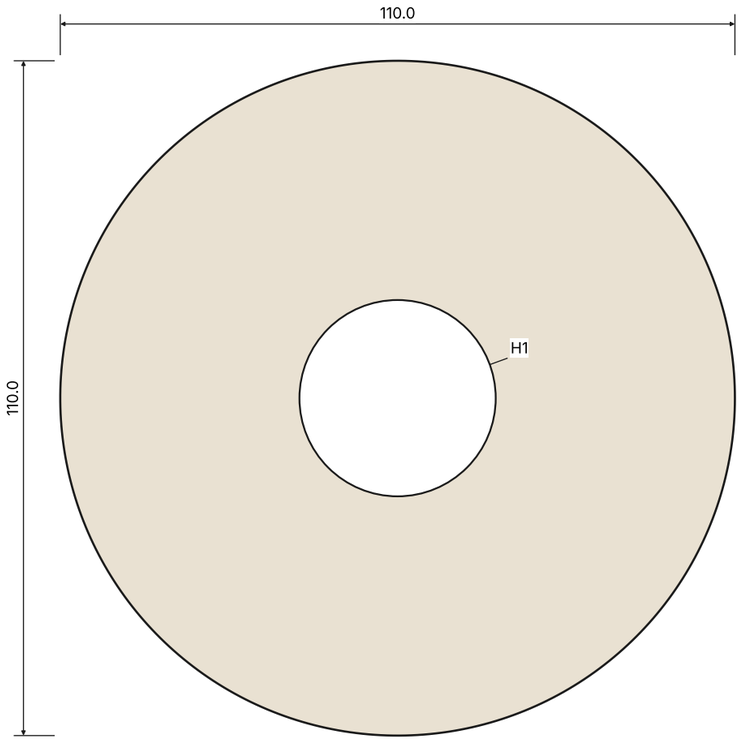
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-05.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-05: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-05

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

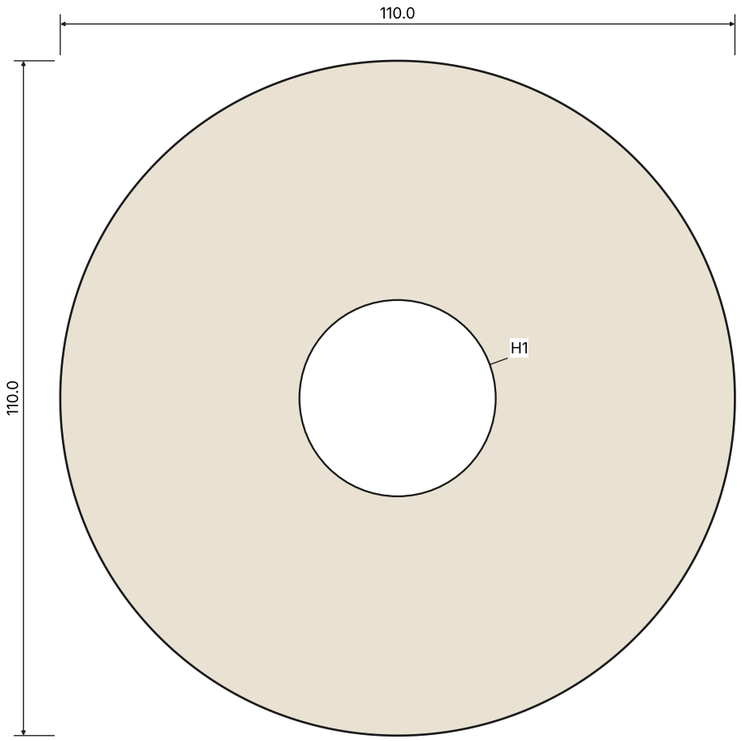
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-06.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-06: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-06

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

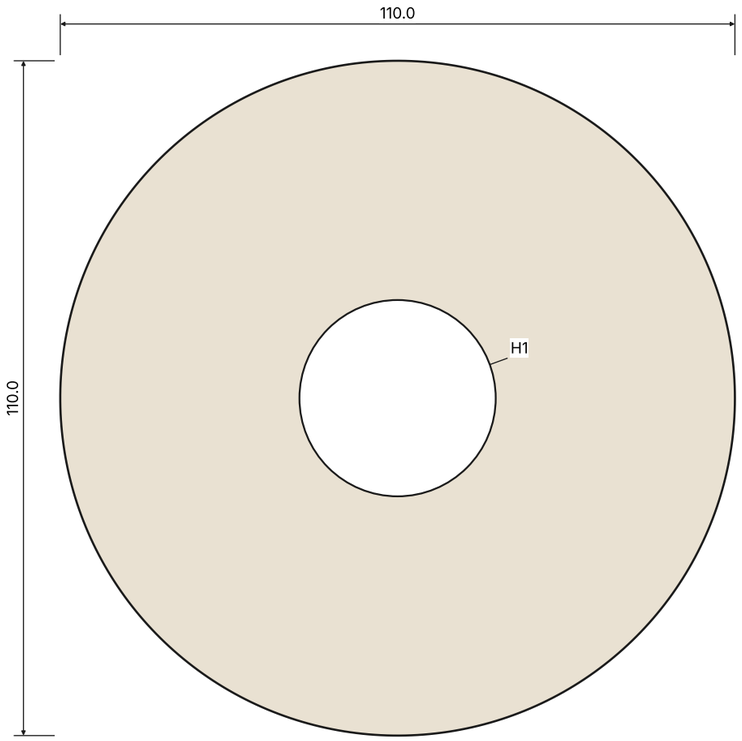
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-07.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-07: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-07

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

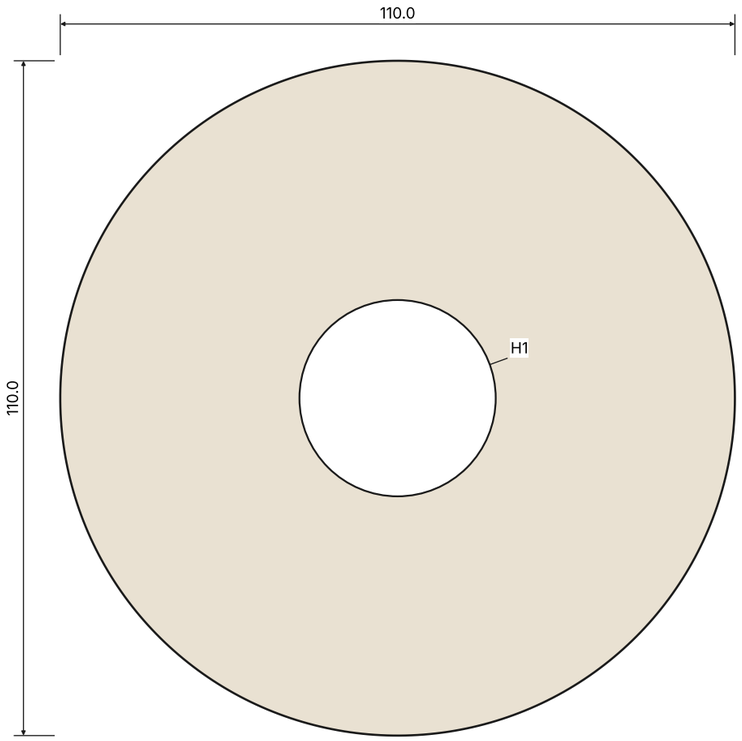
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-08.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-08: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-08

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

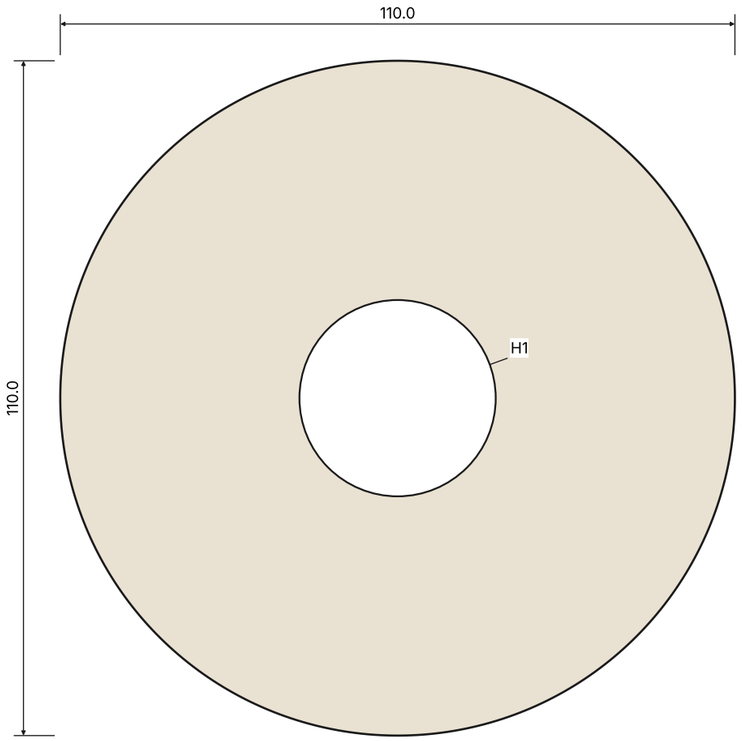
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-09.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-09: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-09

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

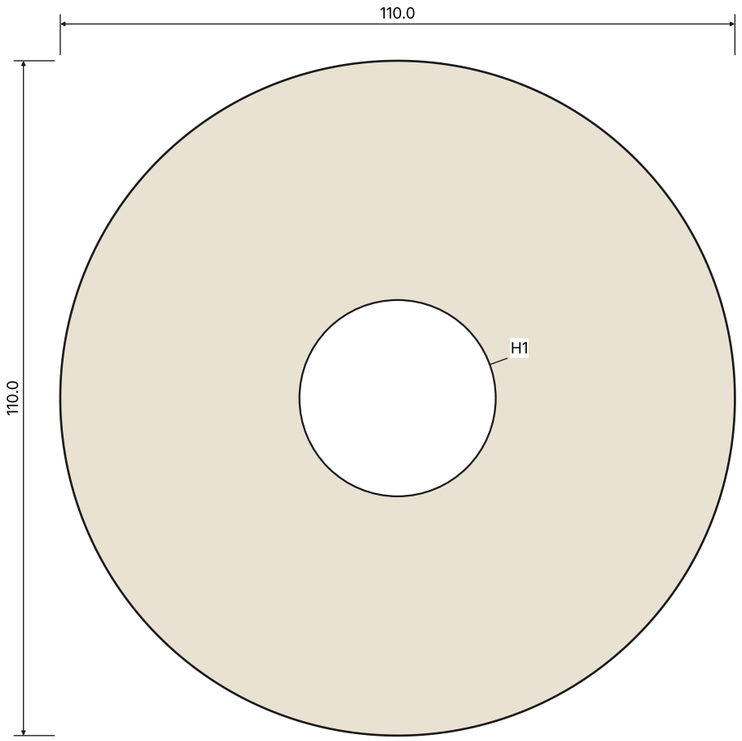
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-10.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-10: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-10

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

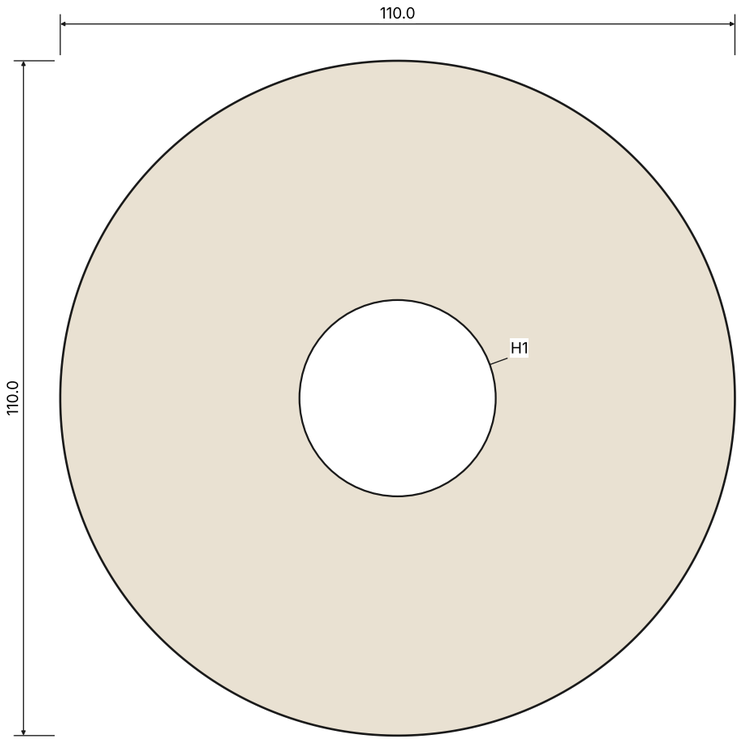
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-11.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-11: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-11

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

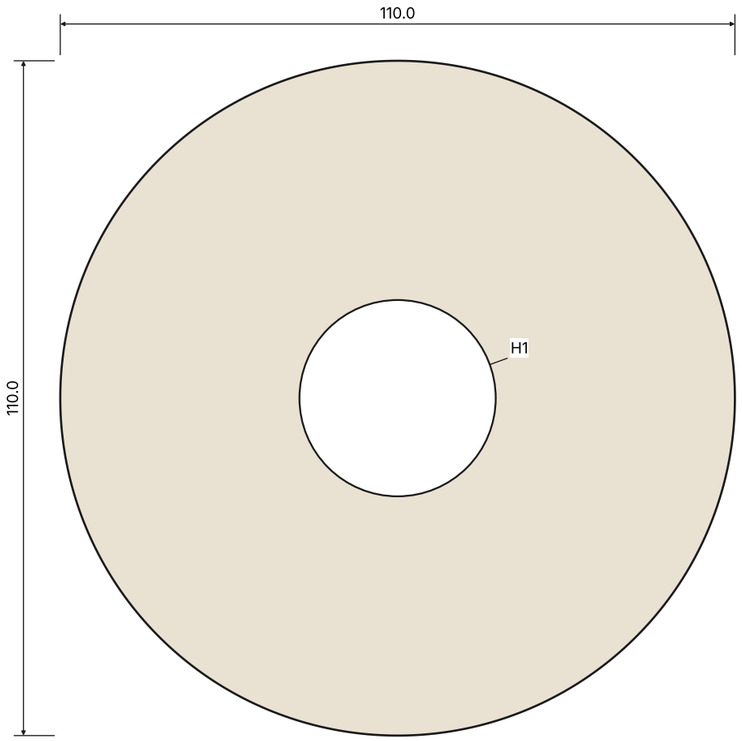
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-12.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-12: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-12

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

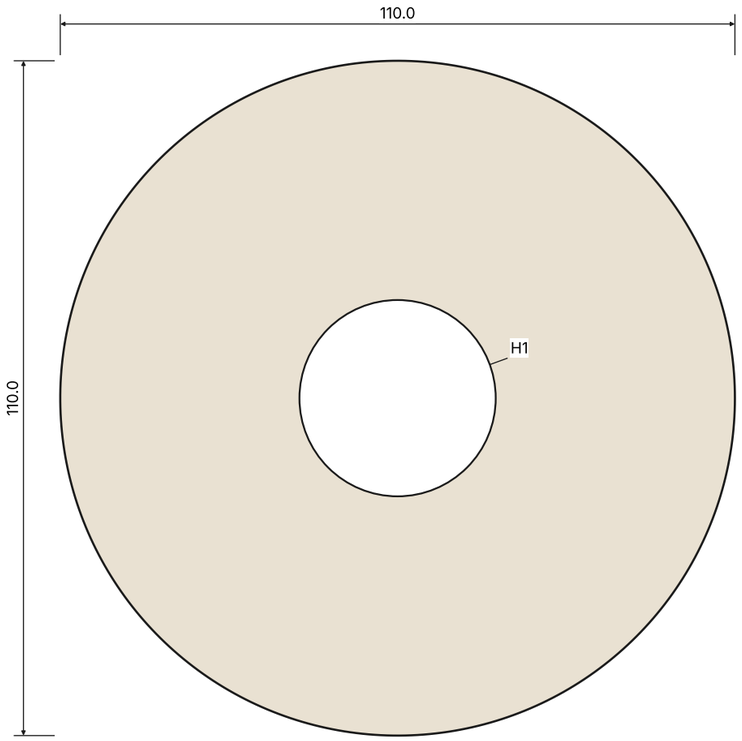
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/disc-13.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø32.0	55.0	55.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-13: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: disc-13

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

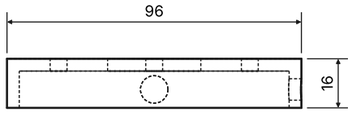
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

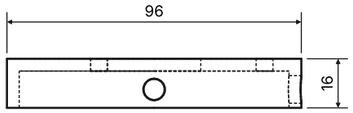
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

4. foot: lathe, aluminium; 39 cm3 each; 1 for the set; 1:2

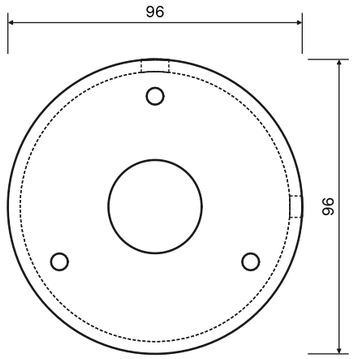
6061 turned from 4 in bar, the pocket and holes milled, bead-blasted, black anodised; the column's three M5 screws go up through its top into the bottom disc's inserts; nylon washers under their heads



FRONT



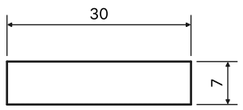
FROM THE RIGHT



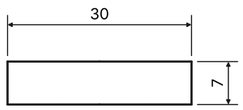
FROM ABOVE

9. plug: lathe, aluminium; 5 cm3 each; 1 for the set; 1:1

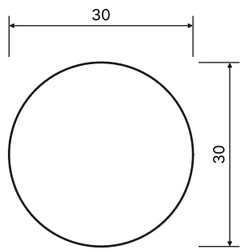
carries the spine (an M4 screw up its centre) and the tube (a 1 mm groove); held in the foot by two M3 set screws, so the whole engine drops out of the bottom once the cover is off



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

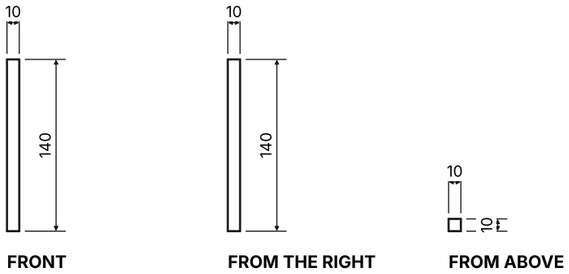
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

10. spine: cnc-mill, aluminium; 14 cm3 each; 1 for the set; 1:5

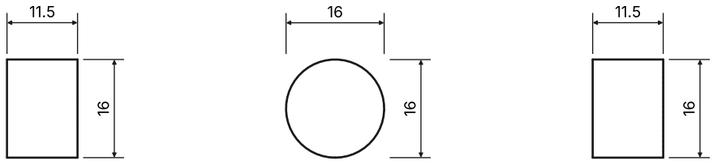
10 mm square 6063 bar, 140 long: the strips' heat path down to the plug and the foot



FRONT **FROM THE RIGHT** **FROM ABOVE**

13. knob: lathe, stainless; 2 cm3 each; 1 for the set; 1:1

304 stainless, turned and diamond-knurled, a set screw onto the dimmer's 6 mm D shaft: turn to dim, click off at the end of its travel



FRONT **FROM THE RIGHT** **FROM ABOVE**

Printed and machined parts

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	disc-01, disc-02, disc-03, disc-04, disc-05, disc-06, disc-07, disc-08, disc-09,	cut	13	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	110 × 110	3.45
2	disc-14	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	110 × 110	3.45
3	inserts	buy	3	any (E-Z LOK, Tappex or alike) Brass threaded insert for plastics, M5, press-in or heat-set, 7 OD x 10 long, knurled		0.60
4	foot	lathe	1	Aluminium 6061	106 × 106 × 26	52.30
5	cover	cut	1	2 mm Aluminium 5052 sheet	87 × 87	1.51
6	bumpers	buy	1	3M Bumpon SJ5302 class clear or black silicone bumpers, 12.7 × 3.5, four		1.00
7	screws	buy	3	M5 × 12 button head socket screws, A2 stainless, black oxide or plain, with nylon washers		1.50
8	cover-screws	buy	4	M3 × 6 countersunk socket screws, A2 stainless, into tapped holes in the foot		1.00
9	plug	lathe	1	Aluminium 6061	40 × 40 × 17	45.68
10	spine	cnc-mill	1	Aluminium 6061	20 × 20 × 150	46.50
11	strips	buy	1	Waveform Lighting (or alike) Dim-to-warm LED strip, 24 V, CRI 95+, 2700 K at full warming toward 1800-2000 K as it dims (CENTRIC WARM class), 8 mm, two runs of 140 mm on the spine's faces, about 4.4 W at full		12.00
12	tube	buy	1	any (Plaskolite or Spartech opal cast tube) White translucent (opal) cast acrylic tube, 1 in OD x 1/8 in wall, cut to 150 and capped with an opal disc bonded in, about 50-60 % transmission		6.00
13	knob	lathe	1	Stainless steel 304	22 × 26 × 26	45.44
14	dimmer	buy	1	Allanson / LEDSupply / Waveform (to choose) A 12-24 V DC PWM LED dimmer board with a rotary potentiometer and end-of-travel switch, PWM 20 kHz or above (flicker-free), out of its housing, on two M3 standoffs in the foot		12.00
15	jack	buy	1	Switchcraft or CUI Panel DC power jack 5.5 × 2.1, threaded M8 bush, with a sealing O-ring		4.00
16	adapter	buy	1	to choose (Class 2 / LPS, DOE Level VI) 24 V 15-18 W desktop adapter, UL 62368-1 listed, Class 2 and LPS on its label, 2.1 mm plug, a 2 m black fabric-covered cord to the lamp		18.00

Notes

1. A proposal (strata/README.md): Core as a core sample. The current Core (orb_table.py) stays the baseline.
2. The column: 14 discs of 1/2 in cast sheet (178 tall at 12.7; the CAD uses 12), 110 across, bonded with polymerising cement (Weld-On 40 class) in a press jig, annealed (80 C, slow cool), turned true, its outside and top diamond-turned and buffed, its bore bored to 32 and polished. About 1.75 kg of acrylic.
3. The light: two runs of dim-to-warm strip (2700 to about 1900 K) on a 10 mm aluminium spine, inside an opal tube, about 4.4 W at full. The engine drops out of the bottom: the cover's four screws, then two set screws.
4. Heat (worked): about 3.3 W of heat down the spine (about 12 K along it) into the foot, 121 cm2 of black anodise: about 27 K over the room. In a 25 C room the spine's top runs about 64 C, and the bore's wall less, against 70 C for the acrylic. To confirm with thermocouples on the first one.
5. Stability: the whole about 2.0 kg, its centre about 101 up; it tips at about 26 deg on its 96 foot.
6. Power: the v1 plug-in (launch/core-v1-plugin.md): a listed 24 V Class 2 adapter into a sealed jack at the back. The battery stays an option: the foot's pocket takes a cell carrier later, the column unchanged.

Parts list and notes

Core Sample: Table lamp: a bored column of bonded tint discs, lit from within

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15